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Lab 08 Bias-Variance Tradeoff
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Reflection on the Bias-Variance Tradeoff

In this lab, our group explored how model complexity affects machine learning performance and why finding the right balance matters. We used a synthetic sine-wave dataset and Polynomial Regression to compare models with degrees 1, 4, and 15. We created learning curves to see how the models performed with different amounts of training data. Although all three models used the same dataset, they behaved differently depending on their complexity. This activity helped us understand underfitting, overfitting, and the bias-variance tradeoff through graphs. We learned that a model can fit its training data very well but still struggle to make accurate predictions on new data.

Polynomial Regression helped us see how changing the degree affects a model's ability to recognize patterns (Tien). We created and trained three models with degrees 1, 4, and 15 using the same dataset (Tien). After training, we made predictions and compared how closely each model followed the data points (Viktoriya). The degree 1 model produced an almost straight line and failed to capture the curved pattern, which showed us an example of underfitting (Viktoriya). The degree 4 model followed the general pattern better, while the degree 15 model created a more complicated curve that followed smaller changes in the data (Zaid). This showed us that increasing complexity can improve predictions, but too much complexity can make it focus on noise instead of the actual pattern (Tien, Zaid, Viktoriya).

One interesting part of Task 1 was seeing how differently the three models behaved despite using the same data (Viktoriya). At first, we thought a more complex model would always make better predictions because it could capture more details (Viktoriya). However, comparing their predictions showed us why that is not always true (Zaid). The degree 15 model followed the training data closely, but its complicated curve raised questions about its performance on new data (Zaid). We also noticed that the degree 1 model could not capture the sine-wave pattern because a straight line was too simple (Tien). This helped us understand why we should consider both model complexity and its ability to recognize the general pattern rather than focusing only on the training points (Tien).

In Task 2, we used learning curves to compare training and cross-validation scores for the underfitting and overfitting models (Tien, Zaid, and Viktoriya). We worked with models of degrees 1 and 15 and examined how their scores changed as we increased the amount of training data (Tien). For degree 1, the training scores remained relatively low, while the cross-validation scores started much lower and gradually improved (Viktoriya). Even though the gap became smaller, both scores remained relatively low, showing that the model still struggled to capture the pattern (Viktoriya). The degree 15 model had relatively high training scores, but its cross-validation scores were much lower, especially with smaller training sets (Zaid). From these

results, we learned that a model can perform well on training data but struggle with new data (Zaid).

Understanding the learning curves was challenging because we had to compare two scores instead of looking at one result (Viktoriya). We noticed that cross-validation scores could become negative, which confused us at first (Zaid). After comparing them with the training scores, we realized that a high training score does not always mean the model performs well overall (Zaid). We also learned that negative R^2 scores mean the model performs worse than predicting the validation data's average value (Tien). Watching how the scores changed with more training examples showed us why learning curves reveal problems that a single score might hide (Viktoriya). This experience taught us to check both training and validation performance before deciding how well a model works (Tien).

The Knowledge Check helped us connect the graphs to high bias and high variance (Tien, Zaid, and Viktoriya). We identified degree 1 as underfitting because it missed the curved pattern, while degree 15 showed overfitting because it followed the training data too closely (Tien, Zaid, Viktoriya). We found that degree 4 provided a better balance between simplicity and complexity (Tien). The degree 1 learning curve showed that adding more training examples did not solve underfitting because the model lacked complexity (Viktoriya). Meanwhile, the degree 15 model had a large gap between training and cross-validation scores, showing high variance (Zaid). These results taught us that different model problems need different solutions, so we should understand the cause before making changes (Zaid).

Moreover, this lab built on what we learned about Model Evaluation in Lab 07, where we explored different ways to check how well a model performs (Tien). In Lab 07, we used Cross-Validation to see how consistently our model performed, while Lab 08 helped us understand why models sometimes struggle by comparing training and cross-validation scores through learning curves (Viktoriya). We learned that higher training scores do not always mean better predictions on new data (Zaid). This made us wonder whether changing the polynomial degree or adding training examples would improve the degree 15 model's cross-validation scores (Zaid). We would like to test more degrees between 4 and 15 to see when the model becomes too complex (Tien). Additionally, we would like to try different noise levels and compare how the models respond (Viktoriya).

In conclusion, this lab helped our group understand why choosing the right level of model complexity is important. Polynomial Regression showed us how simple and complex models behave differently with the same data. Learning curves helped us recognize underfitting and overfitting by comparing training and cross-validation performance. We learned that a model should not only fit training data but also recognize patterns that help it predict new data. Most importantly, this lab changed how we think about model performance because more complexity does not always lead to better results. In future labs, we want to focus more on balancing model complexity, training performance, and predictions on new data.